

{3} Current Electricity

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- * ELECTRIC CURRENT
- * ELECTRIC CURRENTS IN CONDUCTORS
- * OHM' S LAW
- * DRIFT OF ELECTRONS AND THE ORIGIN OF RESISTIVITY
- * LIMITATIONS OF OHM 'S LAW

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* TEMPERATURE DEPENDENCE OF RESISTIVITY

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- * **Charges in motion constitute an electric current.**
- * In the present chapter, we shall study some of the basic laws concerning **steady electric currents**.

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- * The direction of current is taken as the direction of motion +ve charges or opposite to the motion of -ve charge.

Ex 1 : In cosmic rays 0.15 protons/cm²-s are entering the earth's atmosphere. If the radius of the earth is 6400 km, what is the current received by the earth in the form of cosmic rays?

Ex 2: In neon gas discharge tube 2.9×10^{18} Ne^+ ions move to the right through a cross-section of the tube each second, while 1.2×10^{18} electrons move to the left in this time. What is the net electric current in the tube?

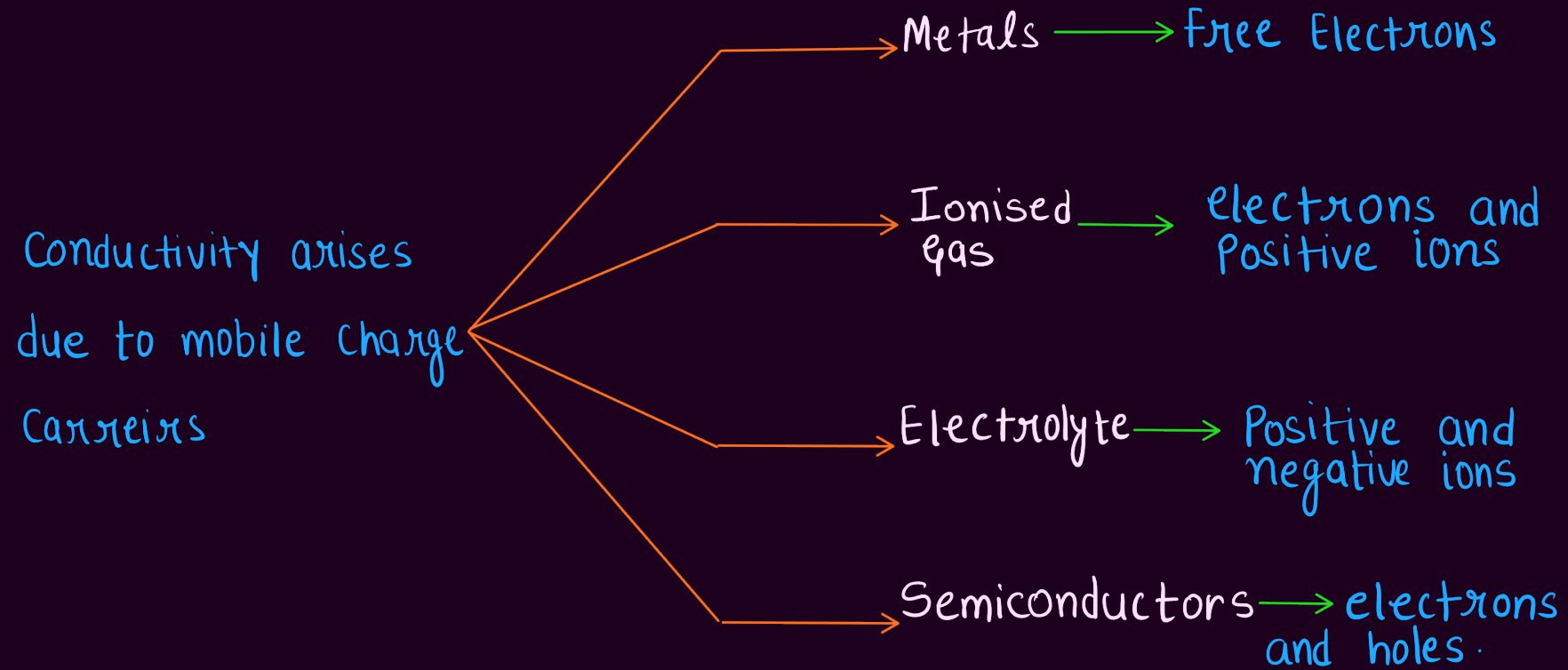
Ex 3 : In the Bohr's model of hydrogen atom, the electron is assumed to rotate in a circular orbit of radius 5×10^{-11} m, at a speed of 2.2×10^6 m/s. What is the current associated with electron motion?

Ex 4 : The current in a wire varies with time according to the relation $I = 2 \text{ A} + (0.6 \text{ A/s}^2) t^2$.

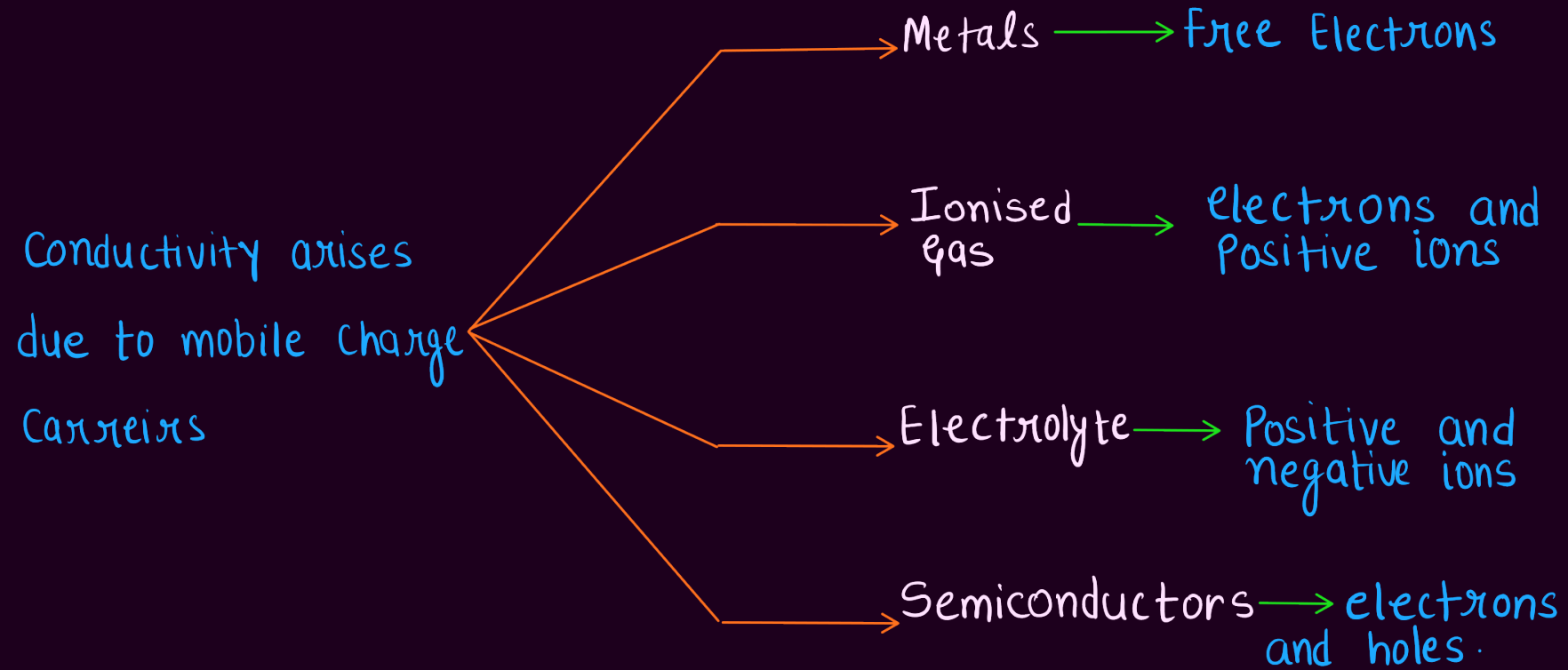
- (a) How many coulomb of charge pass a cross-section of the wire in the time interval between $t = 0$ and $t = 10 \text{ s}$?
- (b) What constant current would transport the same charge in same charge in the same time interval?

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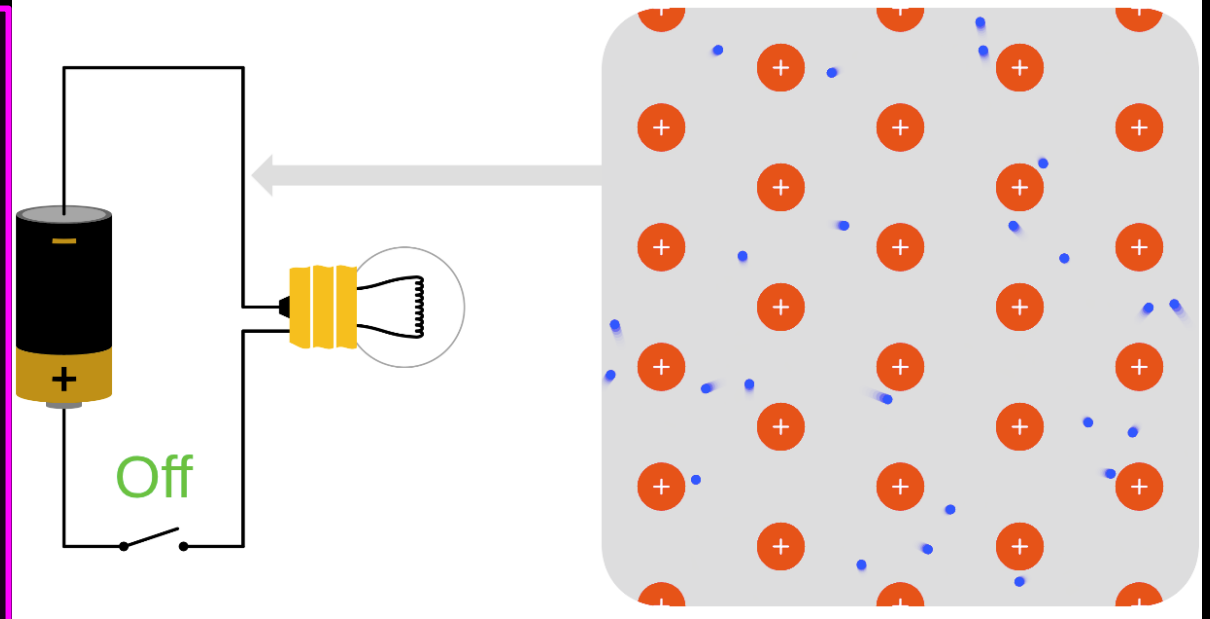


* In our discussions, we will focus only on solid conductors so that the current is carried by the negatively charged electrons in the background of fixed positive ions.

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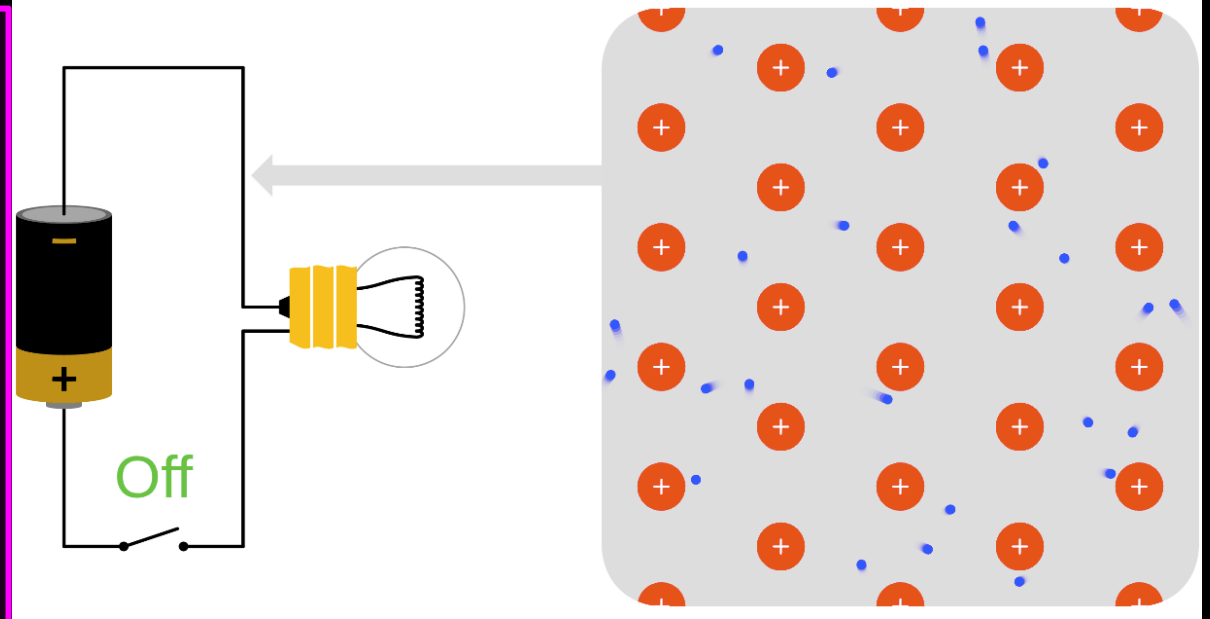
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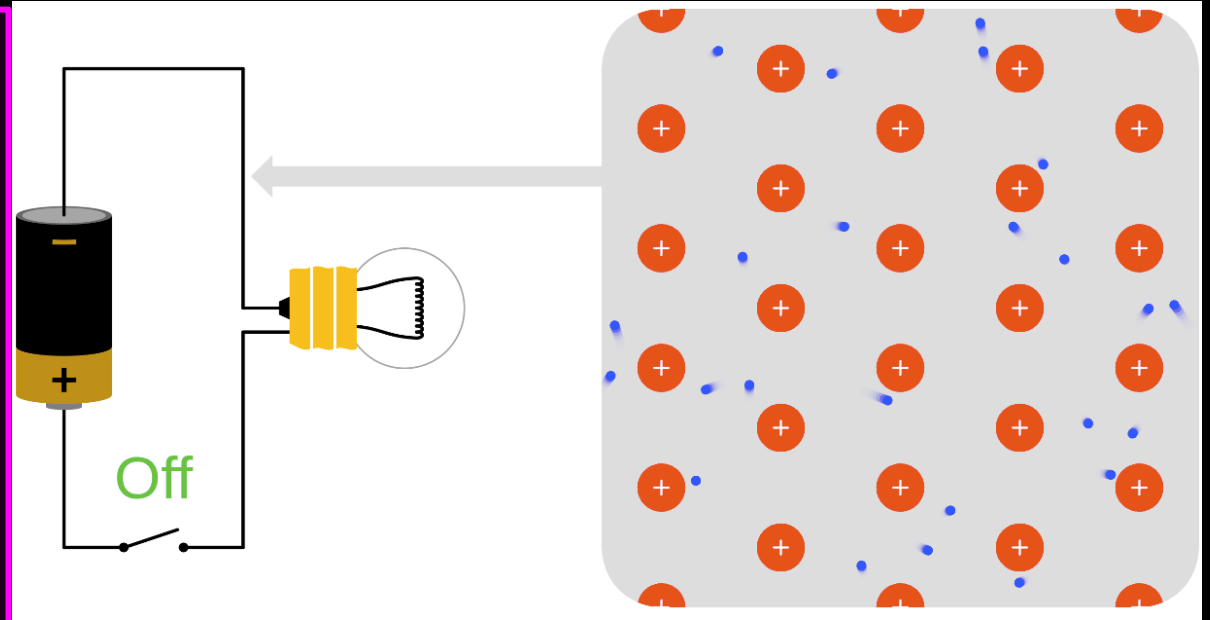
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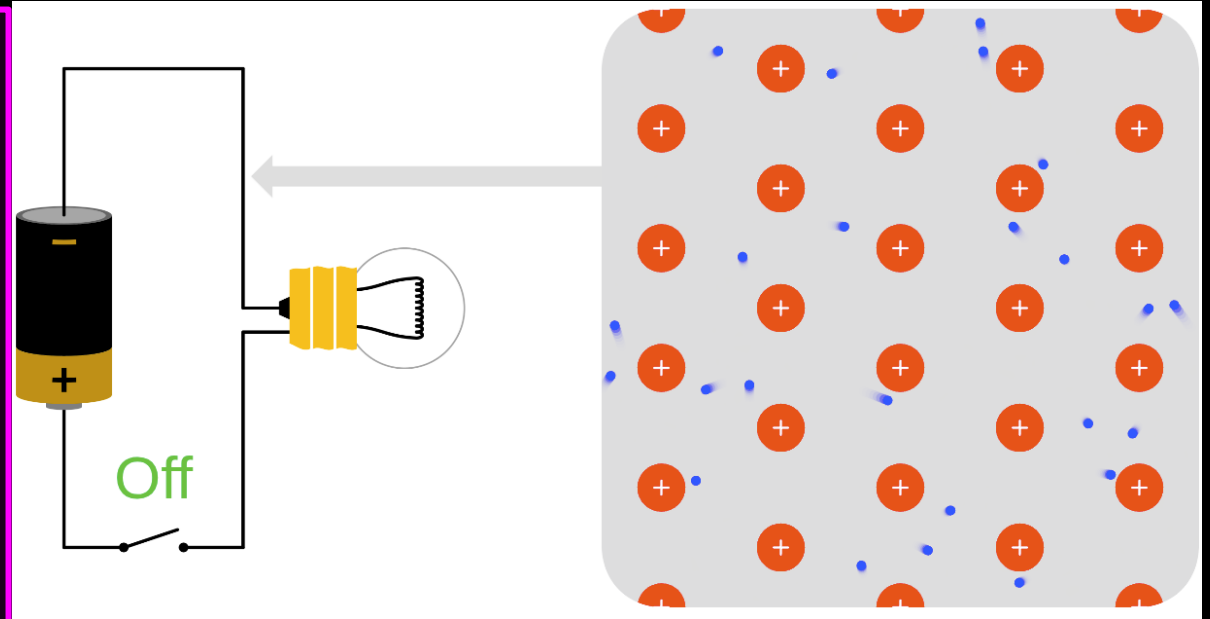


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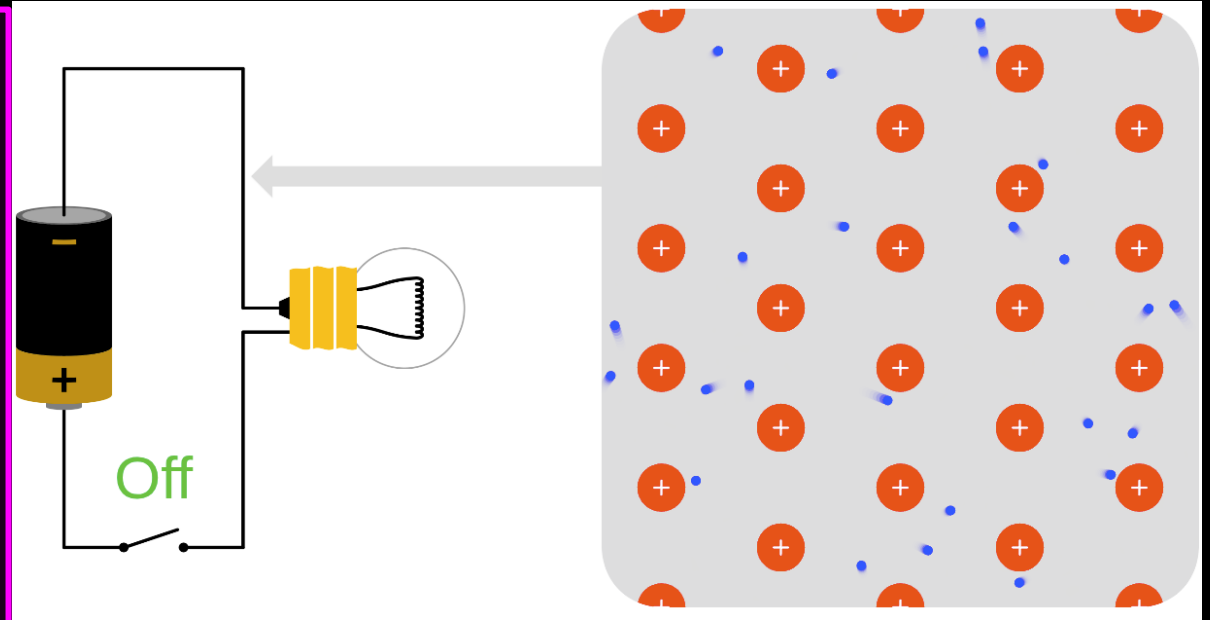


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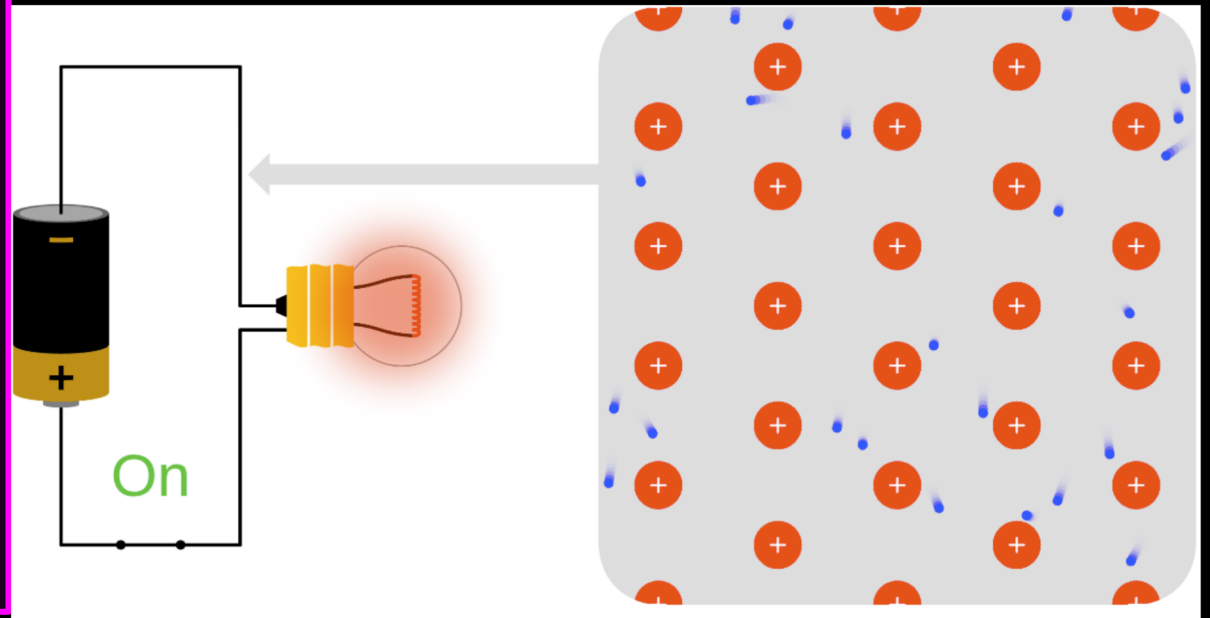
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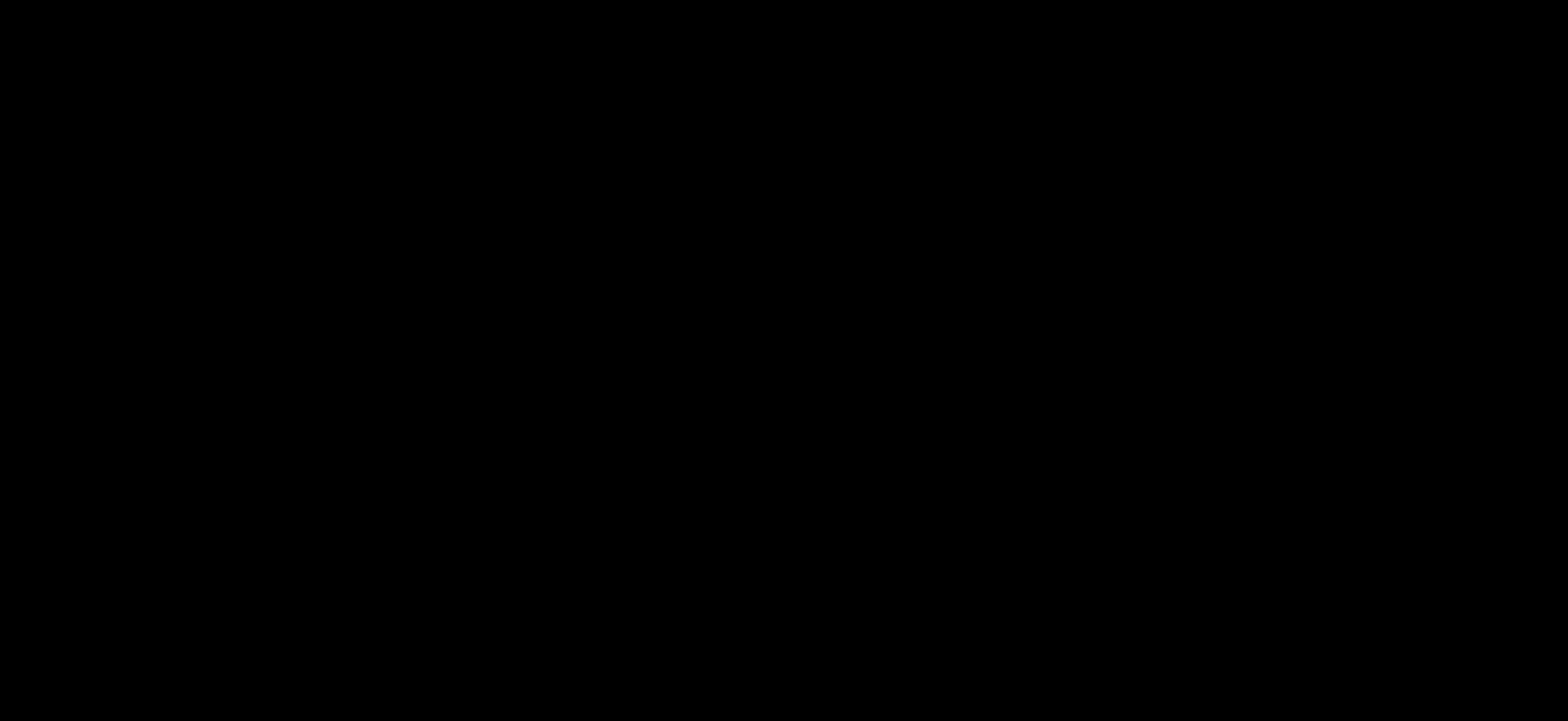


Case 2 : When electric field is present :

* The electrons will be accelerated due to this field, in the opposite direction of the field. The electrons, as long as they are moving, will constitute an electric current.



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$$R \propto \frac{l}{A} \text{ or } R = \rho \frac{l}{A}$$

* where the constant of proportionality called resistivity (ρ) depends on the material

CURRENT DENSITY AND EQUIVALENT FORM OF OHM'S LAW

Consider a rectangular block of material with length l , cross-sectional area A , and resistivity ρ . A potential difference V is applied across the ends of the block, causing a current I to flow.

The electric field E inside the block is uniform and given by $E = \frac{V}{l}$. The current density J is defined as the current per unit area, $J = \frac{I}{A}$.

Ohm's law for the entire block states that $V = IR$, where R is the resistance. The resistance of the block is given by $R = \rho \frac{l}{A}$.

Substituting $V = El$ and $I = JA$ into Ohm's law, we get $El = RJA$. Using the expression for resistance, $R = \rho \frac{l}{A}$, we find $El = \rho \frac{l}{A} JA$.

Simplifying, we obtain $E = \rho J$, which is the equivalent form of Ohm's law in terms of current density.

This equation shows that the electric field is directly proportional to the current density, with the proportionality constant being the resistivity ρ .

For a material with conductivity σ , where $\sigma = \frac{1}{\rho}$, the equation can be written as $J = \sigma E$.

This form of Ohm's law is particularly useful in analyzing the behavior of materials at the microscopic level.

The relationship between current density and electric field is fundamental in understanding the transport properties of materials.

In summary, the equivalent form of Ohm's law in terms of current density is $J = \sigma E$ or $E = \rho J$.

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- * **Current Density (J)** : Current per unit area (taken normal to the current), I/A , is called current density and is denoted by J . It is a vector quantity. The SI units of the current density are A/m^2 .

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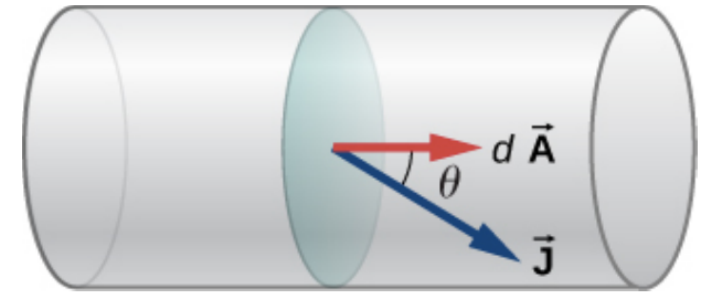


Figure 9.12 The current density $\vec{J}$ is defined as the current passing through an infinitesimal cross-sectional area divided by the area. The direction of the current density is the direction of the net flow of positive charges and the magnitude is equal to the current divided by the infinitesimal area.

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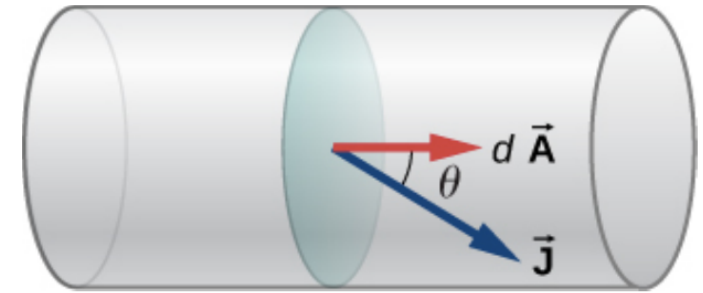


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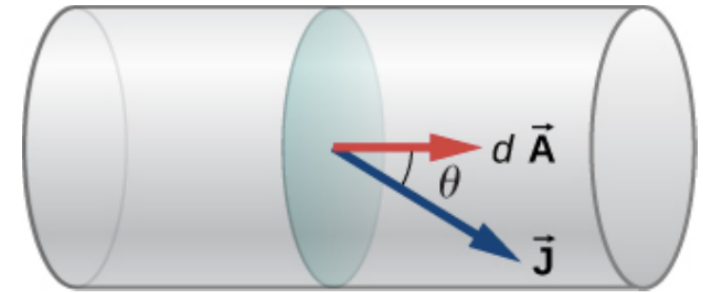


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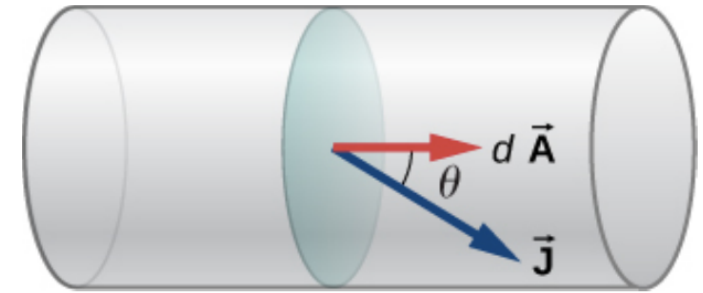


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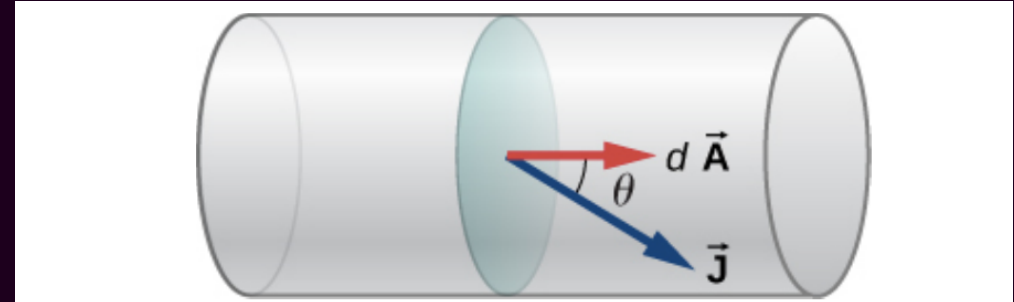


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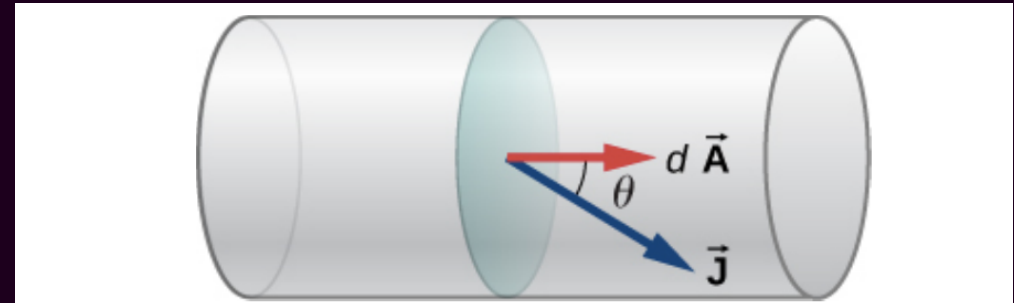


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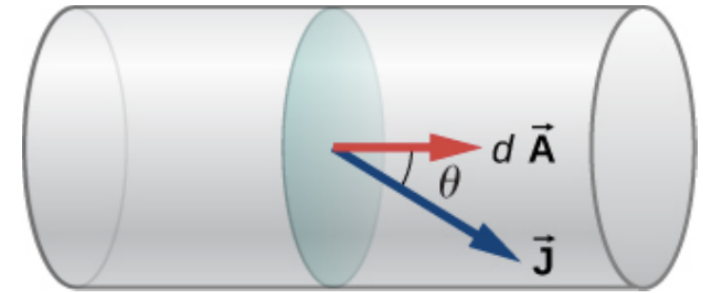


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